Test Case Document: Blockchain-Enabled PDS Beneficiary Registry & Fraud Detection

Target Application: https://demo.vikshitpds.in **Requirement Document:** Blockchain-Enabled Beneficiary Registry Management and Fraud Detection for Public Distribution System (PDS)

1. Overview & Test Scope

This document defines the functional and non-functional test cases required to validate the PDS system against the core requirements specified in the reference document. Testing focuses on verifying beneficiary lifecycle management, fraud detection mechanisms, Aadhaar/e-KYC integration, and the immutability of the blockchain audit trail.

Key Test Modules

1. **Beneficiary Registry & Lifecycle Management**
2. **Fraud & Ghost Beneficiary Detection**
3. **Aadhaar, e-KYC & ePoS Integration**
4. **Blockchain Immutability & Audit Trail**
5. **Multi-Agency Data Synchronization**

2. Test Cases Specification

Module 1: Beneficiary Registry & Lifecycle Management

Test Case ID	Scenario / Description	Pre-conditions	Test Steps	Expected Result
TC_BR_001	Create new valid beneficiary record	User has admin/authorized officer access.	1. Navigate to *Add Beneficiary* page. 2. Enter valid demographic data (Name, DOB, Address, Family Members). 3. Submit the form.	Beneficiary is created with a unique ID, and the transaction hash is generated on the blockchain.
TC_BR_002	Update existing beneficiary demographic details	Beneficiary record exists on system.	1. Search for existing beneficiary. 2. Modify address/economic	Record is updated, prior data remains in historical state, and modification is recorded on the

Test Case ID	Scenario / Description	Pre-conditions	Test Steps	Expected Result
			status. 3. Save updates.	blockchain ledger.
TC_BR_003	Manage beneficiary lifecycle (Family Bifurcation)	Valid family record exists.	1. Select existing family card. 2. Select members to split into a new unit. 3. Confirm split.	Original card updates member count; new family card/ID is generated with linked lineage logged on the blockchain.
TC_BR_004	Mark beneficiary as Deceased / Ineligible	Beneficiary record active.	1. Access beneficiary profile. 2. Mark status as "Deceased" or "Ineligible". 3. Upload death certificate / verification doc. 4. Submit.	Status changes to inactive/deactivated; distribution eligibility is immediately revoked.

Module 2: Fraud & Ghost Beneficiary Detection

Test Case ID	Scenario / Description	Pre-conditions	Test Steps	Expected Result
TC_FD_001	Duplicate Beneficiary Prevention (Aadhaar / Demographic match)	A record with Aadhaar X already exists.	1. Attempt to create a new beneficiary with Aadhaar X. 2. Submit registration.	System flags duplicate record error, rejects creation, and logs the attempt.
TC_FD_002	Flagging Ghost / Deceased Beneficiaries	Death report synced from civil registration database.	1. Run automated fraud detection check or manual verification scan.	System flags ghost/deceased beneficiaries and flags them for removal/verification.
TC_FD_003	Cross-district duplicate record flag	Same demographic/biometric data exists across two districts.	1. Input beneficiary details matching an existing record in a different district/region.	System alerts administrator of multi-district record duplication.

Module 3: Aadhaar, e-KYC & ePoS Integration

Test Case ID	Scenario / Description	Pre-conditions	Test Steps	Expected Result
TC_INT_001	Perform e-KYC	Aadhaar	1. Enter Aadhaar	Verification

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	Authentication during registration	integration sandbox active.	number. 2. Trigger e-KYC verification (OTP / Biometric).	succeeds, matching name and demographic data automatically.
TC_INT_002	Validate ePoS Food Grain Distribution against Registry	Active beneficiary record and valid ePoS terminal session.	1. Initiate grain allocation via ePoS simulation. 2. Authenticate beneficiary via ePoS.	System confirms eligibility, completes distribution, and logs transaction.
TC_INT_003	Attempt distribution to deactivated/ineligible beneficiary via ePoS	Beneficiary status is "Deactivated" or "Ghost".	1. Initiate distribution request at ePoS terminal.	Transaction is blocked by system with an "Ineligible Beneficiary" notice.

Module 4: Blockchain Immutability & Audit Trail

Test Case ID	Scenario / Description	Pre-conditions	Test Steps	Expected Result
TC_BC_001	Audit Trail Verification for Record Creation & Updates	Multiple lifecycle actions performed on a record.	1. Navigate to *Audit Trail / Blockchain Log* section. 2. Search by Beneficiary ID.	Complete time-stamped history of creation, edits, and status changes is displayed with cryptographic transaction IDs.
TC_BC_002	Data Tamper Resistance Check	Record exists in database and blockchain.	1. Simulate unauthorized backend database entry change (if testing environment permits). 2. Run integrity check against blockchain ledger.	System detects discrepancy between centralized store and blockchain, raising a security alert.

Module 5: Multi-Agency Synchronization & Performance

Test Case ID	Scenario / Description	Pre-conditions	Test Steps	Expected Result
TC_SYNC_001	Multi-Agency Data Synchronization	External agency portal integrated.	1. Update status via municipal/external node. 2. Check	Data synchronizes in real time or near real time without data loss or

Test Case ID	Scenario / Description	Pre-conditions	Test Steps	Expected Result
			central PDS registry.	mismatch.
TC_PERF_001	Low Connectivity / Remote Region Offline Sync	Simulated offline ePoS device.	1. Perform transaction offline. 2. Re-establish network connection.	Transaction syncs to blockchain upon reconnect without duplicate entries.